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Commentary

Toward a consensus on the tumor microbiota: Evidence, standards, and interpretation

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Tumors across diverse organs harbor microbial communities that can shape cancer biology and therapeutic responses, yet the field remains polarized by technical and interpretive challenges. In this commentary, we synthesize functional and mechanistic evidence linking intratumoral microbes to cancer hallmarks; critically evaluate current detection approaches; and propose minimal technical and reporting standards to establish microbial presence, viability, and causality. We highlight common pitfalls and outline priorities to move the tumor microbiota field toward robust, clinically actionable insights.

Over the past decade, convergent evidence has established the gut microbiota as fundamental to cancer etiopathogenesis and a key modulator of immunotherapy efficacy across preclinical models and clinical trials. Causality emerged from microbial perturbation through the use of antimicrobials, germ-free/gnotobiotic systems, and human-to-mouse fecal microbiota transplantation (FMT), paving the way for mechanistic dissection of how gut microbes shape host responses to influence both mucosal tumors as well as gut-distal sites (e.g., bone, brain, breast, liver, pancreas, and skin).

The observation that altering the gut microbiota affected tumors outside the intestine demanded explanation: how did intestinal microbes direct intratumoral cellular responses at remote sites? To advance understanding of microbiota-host communication, mechanistic studies addressed three non-exclusive routes: whether (1) microbial structural components—including cell-wall fragments and phage products—reached tumors and conditioned the microenvironment; (2) gut-derived metabolites trafficked systemically, accumulated within tumors, and modulated tumor and immune-cell programs; and (3) viable microbes translocated from the gut (or other mucosal sites) to gut-distal tumors and reprogrammed local immunity. Guided by these questions, functional studies coupled with comprehensive, contamination-aware multi-omics across independent laboratories yielded two key insights. First, they confirmed the accumulation of microbial signals—and, in many cases, viable commensals—within gut-distal tumors. Second, they demonstrated that intratumoral microbes directly reprogram tumor and immune cells, thereby shaping cancer etiopathogenesis and influencing therapy outcomes.

These findings motivated a precise, operational definition of the tumor microbiota. We define the tumor microbiota as the ensemble of microorganisms (bacteria, archaea, fungi, viruses/phages, and protists) as well as their nucleic acids, proteins, and metabolites physically localized within the tumor parenchyma, stroma, and interstitial or vascular spaces—distinct from luminal/surface communities—including viable and non-viable microbes capable of interacting with local host cells.

This commentary synthesizes current evidence on the tumor microbiota and offers practical guidance to avoid misinterpretation. We highlight exemplary studies linking intratumoral microbes to established cancer hallmarks and propose "best-evidence" standards—contamination-conscious study design, orthogonal validation, and functional readouts that probe causality. We conclude with a methods roadmap that compares widely used approaches, delineates their strengths and limitations, and sets criteria for distinguishing causality from correlation. Our focus is on tumor-microbiota data from gut-distal, low-biomass cancers—brain, breast, bone, pancreas, liver, and skin (melanoma)—where claims are potentially high impact yet demand heightened scrutiny.

Impact of intratumor bacteria on cancer hallmarks

Here, we highlight mechanistic studies that define how tumor-associated microbial communities and individual microbes within tumors shape cancer and immune cell responses and modulate sensitivity to cancer therapies. We focus on work that rigorously confirms microbial

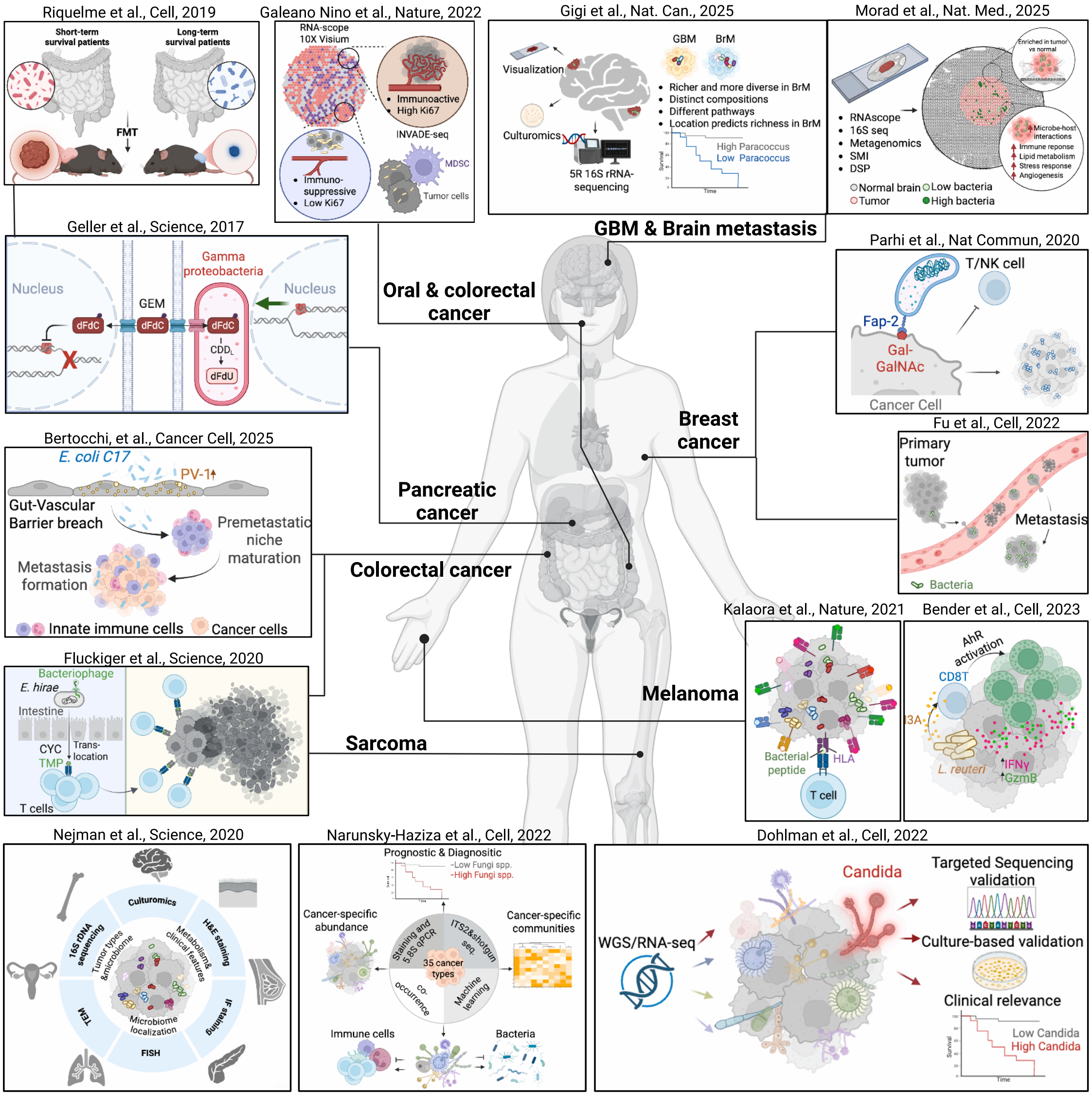

Figure 1. Key mechanistic evidence linking the tumor microbiota to cancer etiopathogenesis

GBM and brain metastasis: Gigi et al. found that brain tumors harbored tumor-type- and location-specific bacterial signatures, with microbial communities and pathways correlating with metastatic behavior and survival. Morad et al. used high-resolution spatial imaging to reveal intracellular bacterial signals within tumor, immune, and stromal cells, correlating with antimicrobial and immunometabolic signatures across regional and cellular scales.

Pancreatic cancer: Geller et al. showed that intratumoral *Gammaproteobacteria* expressing long-form cytidine deaminase (CDD-L) metabolized gemcitabine (GEM), conferring chemoresistance; antibiotic treatment or colonization with CDD-negative strains restored sensitivity. Riquelme et al. revealed that distinct PDAC tumor microbiota signatures associated with long-term survival; fecal microbiota transplantation (FMT) of "long-survivor" tumor microbiota into mice reprogrammed intratumoral immunity and suppressed tumor growth.

Melanoma: Bender et al. showed that *Lactobacillus reuteri* translocated to melanoma produced indole-3-aldehyde (I3A) that activated AhR in CD8 T cells, thereby enhancing Tc1 responses and anti-PD-L1 efficacy. Kalaora et al. demonstrated that intratumoral bacterial peptides presented by HLA class I shaped the antigenic landscape and anti-tumor T cell responses.

Breast cancer: Parhi et al. reported that intratumoral *Fusobacterium nucleatum* colonized mammary tumors via Fap2 binding to Gal-GalNAc and promoted tumor progression. Fu et al. found that intracellular bacteria in breast cancer cells enhanced their survival under mechanical stress in circulation and facilitated metastatic colonization.

Oral and colorectal cancer: Galeano Niño et al. showed that intratumoral microbiota shaped spatial organization and that microbiota manipulation remodeled immune and stromal architecture. Sarcoma and colorectal cancer: Fluckiger et al. showed that cyclophosphamide (CYC)-induced gut barrier disruption allowed systemic translocation of prophage-bearing *Enterococcus hirae*, which produced phage antigens cross-reactive with tumor antigens, thereby enhancing T cell responses and PD-1 blockade.

Colorectal cancer: Bertocchi et al. showed that *E. coli* C17 impaired the gut-vascular barrier (GVB) through a VirF-encoded type III secretion system that up-regulated vesicle-associated protein-1 (PV-1), fostered premetastatic niche maturation, and increased liver metastasis.

Pan-cancer: Nejman et al. generated a pan-cancer atlas of intracellular, tumor-type-specific bacteria in cancer and immune cells, with microbial signatures correlating with clinical features and therapy responses. Narunsky-Haziza et al. found that tumor-resident fungi formed cancer-type-specific communities that

(legend continued on next page)

presence within tumors and/or demonstrates direct effects on host pathways, thereby influencing cancer behavior and/or treatment outcomes (Figure 1). We apologize for not being able to include all existent relevant studies because of space restrictions.

Growing evidence shows that tumors across diverse organs harbor live microbial communities. Large-scale, contamination-controlled analyses have confirmed bacteria and fungi in multiple solid tumors, including those in the breast, lung, ovary, pancreas, bone, brain, and skin.[1–4] These microbes often reside within cancer and immune cells, underscoring their capacity to directly modulate host-cell behavior.[1–11] Beyond mere presence, intratumoral microbes have been implicated in multiple cancer hallmarks, including genomic instability, sustained proliferation, resistance to cell death, metabolic reprogramming, epithelial-mesenchymal transition, invasion and metastasis, angiogenesis, immune evasion, and therapeutic resistance. Although many mechanisms remain incompletely defined, functional studies now provide concrete examples of microbial control over key malignant traits.

Intratumoral microbes can actively promote metastasis and invasion. In breast cancer, intracellular *Staphylococcus* and *Lactobacillus* enhance circulating tumor cell survival and lung colonization by reorganizing the actin cytoskeleton.[7] In colorectal cancer, *Escherichia coli* disrupts the gut-vascular barrier, fostering pre-metastatic niche formation in the liver and recruitment of metastatic cells.[6] Similarly, *Fusobacterium nucleatum*-infected cancer cells show increased local invasion and migratory capacity.[12]

Intratumor bacteria also modulate responses to anticancer therapies. In pancreatic cancer, *Gammaproteobacteria* expressing the long form of cytidine deaminase metabolize gemcitabine into an inactive compound, thereby protecting tumor cells from its cytotoxic activity.[8] In parallel, the composition of the pancreatic tumor microbiota correlates with outcome; higher intratumoral diversity and specific bacterial species associate with CD8 T cell infiltration and long-term survival while human-to-mouse FMT experiments altered the tumor microbiota as well as tumor growth and immune cell infiltration.[11]

Several studies link intratumor microbes to antitumor immunity and immunotherapy responses. Comparing intratumoral fungal communities with matched bacteriomes and immunomes revealed fungal-bacterial–immune clusters associated with patient survival.[3] In melanoma, intratumoral bacterial signatures differed between responders and non-responders to immune checkpoint therapy, suggesting that tumor-resident bacteria may directly influence therapeutic efficacy.[1] Beyond community-level effects, mechanistic studies have demonstrated that individual intratumoral bacteria can directly modulate anti-tumor immunity. For example, *Lactobacillus reuteri* secretes the tryptophan metabolite indole-3-aldehyde, which activates the aryl hydrocarbon receptor in CD8 T cells, enhances their effector function, and improves immune checkpoint inhibitor (ICI) efficacy in melanoma.[5]

Microbial influences on antitumor immunity extend beyond local metabolite signaling. Cyclophosphamide treatment promotes systemic translocation of *Enterococcus hirae* from the gut, enabling presentation of a prophage-encoded epitope that elicits CD8 T cells cross-reactive with tumor antigens; this microbiota-tumor molecular mimicry enhances antitumor immunity and constrains tumor progression.[13] In addition, peptides derived from intracellular bacteria can be displayed on HLA-I and HLA-II molecules of melanoma cells and elicit T cell reactivity, providing a direct route by which intratumoral bacteria shape tumor antigenicity.[9]

Multiple studies further showed that some bacteria preferentially home to tumors after intravenous delivery, with consequent effects on tumor biology.

Taken together, these studies indicate that tumor-resident microbes can remodel both cancer-cell-intrinsic programs and microenvironmental interactions, adding a microbial dimension to classical models of tumor biology. Systematic dissection of these interactions may open new avenues for diagnostic biomarkers and microbe-centered or microbe-informed cancer therapies.

Challenges in the tumor microbiota field

As a rapidly evolving field, cancer-microbiota research faces recurring obstacles that demand standardized, high-rigor approaches. Clearly defining these challenges is critical for developing robust strategies; we highlight the main issues below.

Reproducibility of microbial biomarkers remains a major challenge. Inconsistent sample storage and preparation, batch effects, and substantial differences between analytical pipelines (Bracken vs. MetaPhlan) and their reference genomes all contribute to variability in findings. Attempts to infer microbial reads from transcriptomic datasets have been hampered by limitations in computational pipelines and data interpretation, and defining robust, reproducible microbial biomarkers is further complicated by heterogeneous cohorts heavily confounded by environment, diet, medications, and lifestyle.

Vulnerability of low-biomass specimens to contamination and methodological bias is a major challenge in tumor microbiota research. Contamination can arise from reagents, the environment, and skin flora introduced during handling and in the reference microbial genomes, including human DNA, primarily high-copy human repeats such as Alus, LINEs, and SINEs, as well as vector sequences originating from manufacturer-specific sequencing primers. Shotgun datasets are often dominated by host DNA, making rigorous depletion and filtering essential to avoid spurious microbial calls. Formalin-fixed paraffin embedded (FFPE) tissues are particularly prone to artifactual signal, and downstream computational normalization and filtration can further introduce bias.

Readouts beyond sequencing, via multiple orthogonal approaches, are essential for establishing true microbial

interacted with bacteria and host immunity and that microbial signatures were linked to survival and immune phenotypes. Dohlman et al. reported that fungal DNA and RNA were detected across tumors; distinct mycobiome patterns correlated with immune pathways and clinical outcomes.
AhR, aryl hydrocarbon receptor; MDSC, myeloid-derived suppressor cells; GBM, glioblastoma; BrM, brain metastasis; SMI, spatial molecular imaging; DSP, digital spatial profiling; dFcC, 2′,2′-difluoro-2′-deoxycytidine; dFdU, 2′,2′-difluorodeoxyuridine; CYC, cyclophosphamide; TMP, tail length tape measure protein; PDAC, pancreatic ductal adenocarcinoma.

influence. Standard pipelines provide limited information on microbial viability and spatial context, whereas demonstrating viable tumor microbes—using fresh, antibiotic-free biospecimens—supports their capacity to actively engage the tumor microenvironment. Mapping their cellular and subcellular distribution, including the host cell types and cytoplasmic, nuclear, or perinuclear localization, can reveal how they modulate tumor signaling and mechanisms.

Discriminating local microbial effects is essential for determining whether the observed impact on cancer development is mediated by microbes within the tumor microenvironment or by microbes in the gut and other mucosal sites. Such spatially resolved analyses are critical for establishing causality and for defining when and where the tumor microbiota itself plays a decisive role.

Challenges in causal inference often require resource-intensive microbial colonization and gnotobiotic/murine models. The process of freezing and/or lyophilization of microbes from donors can diminish the recovery of the entire spectrum of original microbial communities. Gnotobiotic mice, particularly germ-free animals, have underdeveloped immune systems with reduced frequencies of key immune components, including Tregs, Th17 cells, and IgA responses, making them suboptimal for studying immune-mediated mechanisms of tumor microbes. Moreover, selectively manipulating the tumor microbiota without perturbing microbial communities at other body sites remains extremely challenging and, for many tumors (particularly those that are anatomically inaccessible), often not feasible. Oral antibiotics are widely used to target tumor-resident microbes but cannot distinguish between effects at the tumor versus the gut or oral cavity, complicating efforts to pinpoint tumor-specific microbial mechanisms.

Delineation of effects of single microbes versus whole communities is crucial for identifying dominant taxa that drive pro-tumorigenic or antitumorigenic effects. Studying fastidious microbes in the laboratory adds further complexity, and introducing a single taxon can disrupt its native interactions with other tumor microbes and alter commensal communities at distant sites. Moreover, community-level functional diversity may underlie tumor modulation, necessitating transfer of intact microbial consortia. Therefore, distinguishing species-specific from community-mediated effects is essential for guiding strategies to manipulate the tumor microbiota in clinical and preclinical settings and for informing the choice of translational models.

Translational validation is arguably the greatest current challenge, as the field moves from providing correlation to mechanistic insights. Progress is limited by the scarcity of longitudinal, deeply annotated clinical datasets that track microbial dynamics alongside outcomes and by the high safety bar and long timelines required for microbiota-modulating clinical trials to advance to later phases. Addressing these gaps will require community standards for sampling and decontamination, spike-in and mock-community controls, rigorous metadata capture, and harmonized bioinformatics to reduce heterogeneity, enhance transparency, and improve reproducibility.

Establishing tumor microbiota presence and function requires orthogonal evidence

Recognition that tumors harbor microbiota has reshaped our understanding of cancer. Demonstrating microbial presence and functional relevance remains technically demanding and requires orthogonal lines of evidence (Figure 2A). Tumor tissues, particularly at gut-distal sites, are low in microbial biomass and rich in host nucleic acids, posing challenges particularly for sequencing-based approaches. Under these conditions, even minimal contamination during sample collection, processing, or library preparation can obscure biological signals. Amplification bias and sequencing artifacts further complicate data interpretation, often producing misleading microbial profiles.

These challenges underscore that no single approach can reliably determine the presence, localization, and function of tumor-resident bacteria (Figure 2B). Techniques relying solely on sequencing or imaging offer partial insights and may yield misleading conclusions. A robust understanding requires orthogonal frameworks integrating complementary methods to validate microbial presence, viability, and activity.

Profiling approaches provide taxonomic and, to some extent, functional insights into the tumor-associated microbiome. Amplicon-based sequencing of the 16S rRNA gene is widely used for bacterial community profiling, enabling taxonomic classification across tumor types.[1] For low-bacterial DNA samples, DNA extraction kits optimized for microbial rather than host DNA should be used.[7] Rigorous positive and negative controls are essential for detecting amplification artifacts and reagent-derived contaminants. Beyond taxonomic profiling, shotgun metagenomic sequencing offers unbiased, strain-level resolution and enables functional characterization of microbial genes and pathways. However, its application to tumor specimens is hampered by excess host DNA and suboptimal recovery of bacterial DNA due to inefficient lysis of microbial cell walls. Successful implementation typically requires host DNA depletion, optimized extraction protocols, and increased sequencing depth. Across both 16S and shotgun metagenomic approaches, focusing on higher-order community signatures rather than individual microbes can help mitigate the noise inherent in single-microbe identification from large-scale, low-biomass datasets. Microbial reads can also be mined from public human genomic and transcriptomic datasets. For example, the recently developed computational framework PRISM (precise identification of species of the microbiome) enables robust detection of microbial signals from diverse human genomic resources, including The Cancer Genome Atlas (TCGA) and Clinical Proteomic Tumor Analysis Consortium (CPTAC).[14] These approaches provide rapid, large-scale, and cost-effective access to patient material. However, sequencing methods in these resources were often optimized for human rather than microbial nucleic acids—most notably poly(A)-enrichment strategies and host-focused extraction and library preparation—which bias against microbial detection, complicate interpretation, and increase the risk of artifactual signals from contamination or batch effects. Consequently, while poorly suited for comprehensive microbiome profiling, such datasets can still be leveraged for targeted analyses of specific strains, genes, or pathways when processed with rigorous

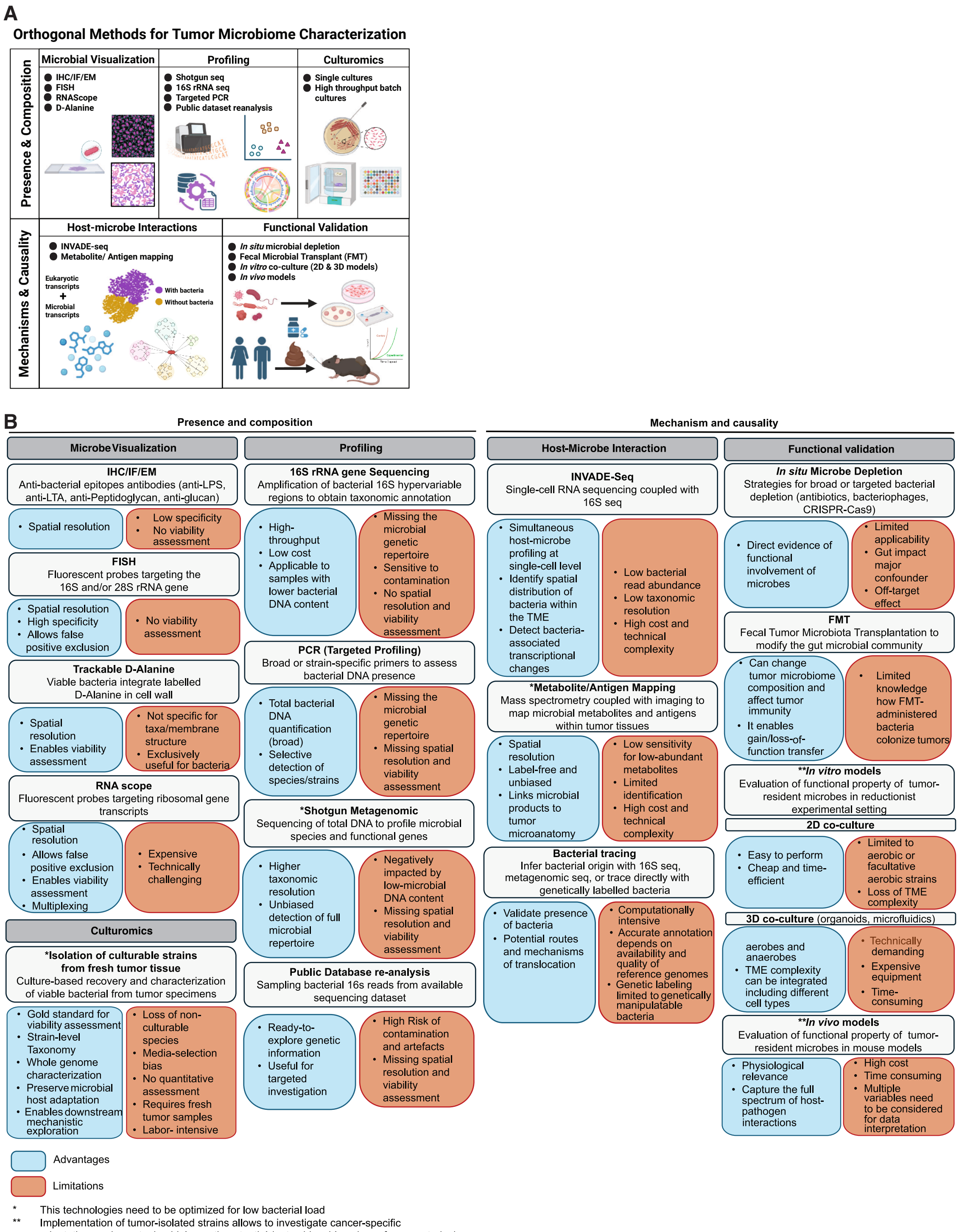

Figure 2. Orthogonal methodologies to establish tumor microbiota presence and function

(A) Schematic overview of orthogonal methods for tumor microbiota characterization. The upper panel highlights complementary approaches to define microbial presence, spatial distribution, and composition, including visualization-based methods (e.g., immunohistochemistry [IHC], immunofluorescence [IF], electron microscopy [EM], fluorescence *in situ* hybridization [FISH], RNAscope, and metabolic labeling) and sequencing-based profiling (16S rRNA gene sequencing, shotgun metagenomics, and RNA-seq) to resolve taxonomic composition and functional potential and culturomics to provide direct evidence of microbial viability and enable downstream mechanistic studies. The lower panel summarizes methods designed to interrogate mechanisms and causality, including host-microbe interaction assays and functional validation approaches, such as fecal microbiota transplantation (FMT), alongside emerging single-cell and spatial modalities (e.g., INVADE-seq [invasion-adhesion-directed expression sequencing] and spatial metabolomics) that integrate bacterial detection with host transcriptional or metabolic states to link microbial presence to functional consequences within the tumor microenvironment. (B) Orthogonal approaches to study the tumor microbiota, spanning microbial presence /composition and functional relevance with associated advantages and limitations.

bioinformatic denoising and decontamination pipelines.

Bacterial visualization provides direct evidence of microbial presence within tissues. PCR-based approaches detect bacterial DNA, including strain-specific signatures, but do not report on spatial distribution or cell viability. In contrast, imaging techniques such as immunohistochemistry (IHC), immunofluorescence (IF), fluorescence *in situ* hybridization (FISH), and electron microscopy (EM) resolve the spatial localization of bacteria within tissues. EM enables ultrastructural, direct visualization of bacteria and can be paired with immunogold labeling (e.g., anti-LPS antibodies) to increase specificity.[4] FISH using probes targeting bacterial 16S rRNA improves taxonomic resolution and spatial accuracy but—like other imaging approaches—cannot distinguish viable from non-viable bacteria.[1] Conversely, RNAscope targeting bacterial rRNA transcripts identifies transcriptionally active bacteria, reducing false-positive signals from dead cells.[12] Similarly, metabolic labeling approaches, such as fluorescent D-alanine incorporation, enable visualization of metabolically active bacteria *in situ*, as fluorescent D-amino acids are incorporated into peptidoglycan during cell wall synthesis.[1,4,15] Collectively, these imaging-based techniques confirm not only bacterial presence but also localization and, in some cases, viability within the tumor compartment, thereby excluding potential environmental contaminants.

Culturomics and related culture-based approaches provide direct evidence of microbial viability and enable functional validation, and—when performed rigorously with appropriate negative controls—they remain the gold standard for confirming that microbes are alive within tumor tissues. Although limited to cultivable taxa and susceptible to biases from microbial competition, differences between solid tissue and liquid media, and the presence or absence of host metabolites, culture-based approaches provide an indispensable platform for mechanistic investigation. Isolation of viable strains enables functional studies, including genomic, proteomic, and metabolomic characterization. To detect low-biomass or intracellular bacteria, culture approaches can be adapted with enrichment in liquid media to improve isolation, although this may favor fast-growing species and reduce diversity. Compared to commercial strains, tumor-isolated bacteria provide a more relevant model. When integrated into preclinical systems, such as *in vitro* co-cultures, microfluidic chips, or *in vivo* models, they facilitate investigation of bacterial properties and their interactions with host tissues, supporting a more accurate understanding of tumor-microbiota dynamics. Complementary strategies, including bacterial depletion with broad or targeted approaches, provide additional support for bacterial causality.

Spatial and advanced imaging techniques dissect host-microbe interactions at single-cell and spatial resolution. These techniques validate microbial presence, define micro-niches, and analyze spatial co-localization with host or immune cells, providing key mechanistic insights and inference in low-biomass contexts. INVADE-Seq,[10] which couples single-cell RNA sequencing (RNA-seq) with bacterial read detection, has revealed intracellular bacteria within tumor and immune cells, highlighting microbial influence on host transcriptional states.[12] Spatial metabolomics maps metabolite distributions within tumor sections, uncovering bacterial footprints.[5,15] A newly developed bacterial-MERFISH method enables profiling of single bacterial cells within complex gut environments. Although its current applications are limited to cultured samples and healthy tissues, this approach has the potential to deliver single-cell-resolution transcriptional profiling of both bacterial and host cells within tumors. These approaches bridge the gap between microbial presence and functional consequence within the tumor microenvironment. Importantly, microbial sequences detected even by multiple methodologies do not equate to viable microbes and may represent microbial signals that require further functional validation.

Bacteria tracing approaches provide insights into the origin of the intratumoral microbiota and serve as secondary validation tools while offering mechanistic understanding. The source of tumor-associated bacteria can be inferred using 16S rRNA sequencing or metagenomic data, including analyses of single nucleotide variants and oligotyping. Additionally, genetically labeled bacteria can be used to trace bacterial lineage dissemination and tumor colonization. In future work, bacterial tracing strategies will be critical for defining the provenance and trafficking routes of intratumoral microbes while also providing orthogonal validation of low-biomass signals.

Together, these orthogonal approaches provide a framework for studying the tumor microbiota as an integrated system (Figures 2A and 2B). Visualization anchors microbial presence, spatial localization, and viability; sequencing defines taxonomy and genetic repertoire; and culturomics confirms viability and enables experimental models to interrogate function. As the field matures, such rigor will be essential for translating microbiota discoveries into actionable, clinically meaningful insights in oncology.

Future directions

Convergent evidence from independent laboratories worldwide, using visualization, sequencing, culture, and functional perturbation, strongly supports a direct role of tumor microbes in modulating cancer hallmarks and therapy responses. While demonstrating the presence of intratumoral microbes demands orthogonal, contamination-controlled approaches, their absence—and thereby exclusion from cancer etiopathogenesis—should be held to the same rigorous standards. Claims derived from single-modality or repurposed datasets, particularly human genomic resources not optimized for microbial detection, should be interpreted with caution, given the risk of contamination, amplification bias, and misclassification.

Going forward, we argue that the field should move beyond binary "presence/absence" debates toward mechanism-focused and clinically oriented questions—when, where, and whether total bacteria burden or specific taxa matters; how microbes and their metabolites shape tumor and immune programs; what the origins and trafficking routes of intratumoral bacteria are (gut, oral, or other) and whether they seed tumors early or late during tumorigenesis; and ultimately, which patients are most likely to be clinically impacted by these interactions.

Another important area for future investigation is whether eukaryotic microbiota members (e.g., protists) and/or their metabolites are present, viable, and functionally active within extra-intestinal tumors and whether they shape local immune and tumor programs.

We envision multiple paths toward clinical translation. Tumor microbiota features may ultimately serve as diagnostic, predictive, and prognostic biomarkers, informing tumor classification, risk stratification, and selection of therapy, including immunotherapy. Realizing this potential will require continued mechanistic dissection of how intratumoral microbes and their metabolites modulate tumor and immune programs and shape responses to therapy. Such insights should enable rational interventions, including targeted depletion or enrichment of intratumoral taxa (e.g., narrow-spectrum antimicrobials and bacteriophages), delivery of beneficial microbes or defined consortia (including next-generation probiotics), and metabolite-based approaches that emulate microbial functions. Looking ahead, tumor-colonizing microbes themselves may be engineered as programmable vectors to deliver therapeutic cargo directly within the tumor microenvironment. In addition, defining therapeutically relevant microbial communities could inform FMT "super-donor" selection, identify modifiable lifestyle factors, and guide targeted microbial modulation strategies.

Ultimately, progress toward clinical translation will depend on harmonized standards for sampling and contamination control, optimized recovery of low-biomass and intracellular taxa, transparent reporting, and thoughtfully designed clinical trials with companion diagnostics—together enabling robust biomarkers and actionable therapeutic strategies.

AUTHOR CONTRIBUTIONS

All authors contributed to the conception and writing of the commentary.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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